Chapter 3. Permutation Tests

Randomization and population models, conditional inference, permutation tests for common problems

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Overview

A **permutation test** estimates the null distribution of a test statistic by **re-randomization**; it was proposed by Fisher and Pitman in the 1930s. Re-randomization requires a great many repetitions, so the method only became prominent recently, with fast computers. The English word "permutation" in permutation test means mathematically an ordered arrangement, but also "to shuffle" (permute). Strictly speaking "shuffle test" would be more precise and less misleading, but "permutation test" is established and is used here.

3.1 Data-generating models

Permutation tests apply in two probability models: the **randomization model**, which chooses at random the subjects (people or other units) receiving a treatment of interest, and the **population model**, which samples subjects at random from populations. We focus on the two-sample situation.

Example 3.1 (Randomization model)

A new treatment for recovery after surgery is compared with the conventional one.[a] Of $N = 7$ patients scheduled for heart surgery at a Seoul hospital this March, $m = 4$ were chosen at random to receive the new treatment and the remaining $n = 3$ the conventional one. The new-treatment patients were discharged after full recovery on days $19, 22, 24, 26$, and the conventional patients on days $23, 33, 40$. Test H_0: "there is no difference between the new and conventional treatments" against H_1: "the new treatment shortens recovery time."

[a] Example adapted from Ernst (2004).

Example 3.2 (Population model)

We compare the antenna lengths of two subspecies of the tsetse fly, the forest-dwelling *Glossina fusca* and the river-dwelling *Glossina palpalis*. Nine forest flies and six river flies were caught and their antenna lengths measured; the flies can be regarded as random samples from each population:

forest flies: 1.24, 1.38, 1.36, 1.40, 1.38, 1.48, 1.38, 1.54, 1.56
river flies: 1.14, 1.20, 1.30, 1.26, 1.28, 1.18

Test H_0: "there is no difference in antenna length (distribution) between the two species" against H_1: "the antenna lengths differ."

3.2 The randomization model

In Example 3.1, let $X_1, \ldots, X_m$ be the recovery times of the m patients receiving the new treatment and $Y_1, \ldots, Y_n$ those of the n conventional patients. We want a test rejecting H_0 when the difference of mean recovery times $\bar{Y}_n - \bar{X}_m$ is large: with $D = \bar{Y}_n - \bar{X}_m$ as test statistic, we measure how extreme the observed D_{obs} is relative to the null distribution of D.

In the randomization model, the **random assignment** of treatments is the basis of inference. In Example 3.1 the N patients were fixed before the random assignment of treatments. If H_0 is true (the two treatments have the same effect), each patient's recovery time is the same whatever treatment the patient receives. E.g. the first new-treatment patient (19 days) would also have needed 19 days under the conventional treatment. In the randomization model the patients are fixed in advance, so **the observations are not random; only the treatment they receive is**

random (because of random assignment). The assignment that m specific patients out of N get the new treatment and the other n the conventional one is just one of $\binom{N}{m}$ equally likely random assignments.[1] Each assignment occurs with probability $1/\binom{N}{m}$, so the null distribution of D is the distribution of the differences D arising under the $\binom{N}{m}$ assignments.

In Example 3.1, the table below lists D for all $\binom{7}{4} = 35$ possible assignments. Row 5 is the observed difference $D_{\text{obs}} = 9.25$. Under H_0 the null distribution of D puts probability 1/35 on each of the 35 values (Figure 3.1 of the book shows its histogram and dot plot). The null distribution of a test statistic obtained in this way is called the **randomization distribution**.

	new				old			D		new				old			D
1	19	22	23	24	26	33	40	11.000	19	19	24	33	40	22	23	26	−5.333
2	19	22	23	26	24	33	40	9.833	20	19	26	33	40	22	23	24	−6.500
3	19	22	23	33	24	26	40	5.750	21	22	23	24	26	19	33	40	6.917
4	19	22	23	40	24	26	33	1.667	22	22	23	24	33	19	26	40	2.833
5	19	22	24	26	23	33	40	9.250	23	22	23	24	40	19	26	33	−1.250
6	19	22	24	33	23	26	40	5.167	24	22	23	26	33	19	24	40	1.667
7	19	22	24	40	23	26	33	1.083	25	22	23	26	40	19	24	33	−2.417
8	19	22	26	33	23	24	40	4.000	26	22	23	33	40	19	24	26	−6.500
9	19	22	26	40	23	24	33	−0.083	27	22	24	26	33	19	23	40	1.083
10	19	22	33	40	23	24	26	−4.167	28	22	24	26	40	19	23	33	−3.000
11	19	23	24	26	22	33	40	8.667	29	22	24	33	40	19	23	26	−7.083
12	19	23	24	33	22	26	40	4.583	30	22	26	33	40	19	23	24	−8.250
13	19	23	24	40	22	26	33	0.500	31	23	24	26	33	19	22	40	0.500
14	19	23	26	33	22	24	40	3.417	32	23	24	26	40	19	22	33	−3.583
15	19	23	26	40	22	24	33	−0.667	33	23	24	33	40	19	22	26	−7.667
16	19	23	33	40	22	24	26	−4.750	34	23	26	33	40	19	22	24	−8.833
17	19	24	26	33	22	23	40	2.833	35	24	26	33	40	19	22	23	−9.417
18	19	24	26	40	22	23	33	−1.250									

Table 3.2: all random assignments possible in Example 3.1.

The observed $D_{\text{obs}} = 9.25$ is relatively large (far right) in the null distribution. Under H_0 the probability that D is at least 9.25 is $3/35 \approx 0.08$, fairly small, so there is some evidence that the two treatments differ.

This method is called the **randomization test**. To obtain its null distribution, the randomization distribution, we shuffle the given treatment assignment; so it can also be called a shuffle test or permutation test.

p-value and type I error of the randomization test. In the randomization model the p-value of the observed statistic D_{obs} is the probability, under H_0, of a value of D at least as extreme as D_{obs}; "extreme" is determined by H_1. For H_1: "the new treatment shortens recovery," larger D is more extreme under H_0. (If H_1 were "the new treatment lengthens recovery," smaller, i.e. more negative, D would be more extreme.) In the one-sided test of Example 3.1, $D \geq D_{\text{obs}}$ is extreme, and under H_0 the null distribution of D is the discrete uniform distribution on the $\binom{N}{m}$ values d_i, so

$$p = P(D \geq D_{\text{obs}} \mid H_0) = \frac{\sum_{i=1}^{\binom{N}{m}} I(d_i \geq D_{\text{obs}})}{\binom{N}{m}}. \tag{3.1}$$

From the table, 3 of the d_i are at least D_{obs}, so $p = 3/35$.

[1] The binomial coefficient $\binom{N}{m}$ ("N choose m") is the number of ways of choosing m of N objects without order: $\binom{N}{m} = \frac{N!}{m!(N-m)!}$.

By (3.1), p-values are multiples of $1/\binom{N}{m}$. If the d_i have no ties, the null distribution of p is the discrete uniform distribution on $\{1/\binom{N}{m}, 2/\binom{N}{m}, \ldots, 1\}$, each value with probability $1/\binom{N}{m}$. At significance level $\alpha = k/\binom{N}{m}$ the type I error of the randomization test is

$$P(\text{reject } H_0 \mid H_0) = P(p \le \alpha \mid H_0) = P\left(\sum_{i=1}^{\binom{N}{m}} I(d_i \ge D_{\text{obs}}) \le k \;\middle|\; H_0 \right) = \frac{k}{\binom{N}{m}} = \alpha, \tag{3.2}$$

so the type I error is controlled exactly at level α. In this sense the randomization test is called an *exact inference*.

Proof of (3.2) (supplied). Under H_0 the multiset $\{d_1, \ldots, d_{\binom{N}{m}}\}$ does not depend on which assignment was realized, and the realized assignment is uniform over the $\binom{N}{m}$ assignments, so D_{obs} is uniform over the d_i. With no ties, let $d_{(1)} > d_{(2)} > \ldots$; if $D_{\text{obs}} = d_{(j)}$ then $\sum_i I(d_i \ge D_{\text{obs}}) = j$. This count is at most k exactly when D_{obs} is one of the k largest values, which has probability $k/\binom{N}{m}$. ■

Example 3.3

In Example 3.1, what is the p-value if H_1 is "the new treatment lengthens recovery time"? This is lower-tailed, so

$$p = P(D \le D_{\text{obs}} \mid H_0) = \frac{\sum_{i=1}^{\binom{N}{m}} I(d_i \le D_{\text{obs}})}{\binom{N}{m}} \tag{3.3}$$

is 33/35.

In two-sided tests the meaning of "extreme" needs care. Whether D is positive or negative, larger $|D|$ favors H_1, so if H_0 is rejected when $|D| = |\bar{Y}_n - \bar{X}_m|$ is large,

$$p = P(|D| \ge |D_{\text{obs}}|).$$

The randomization distribution is not symmetric about 0, so doubling a one-sided p-value is not exact: $P(|D| \ge |D_{\text{obs}}|) \ne 2\min\{P(D \ge D_{\text{obs}}), P(D \le D_{\text{obs}})\}$ in general.

Example 3.4

In Example 3.1, let H_1 be "the new treatment's recovery time differs from the conventional one." The upper-tailed p-value is 3/35 and the lower-tailed one 33/35; the two-sided p-value is 4/35.

Test statistics of the randomization test. So far we used $D = \bar{Y}_n - \bar{X}_m$. Why the difference of the two treatment means? Would another statistic, e.g. a randomization test based on the two-sample t-statistic, be more efficient? The efficiency of a randomization or permutation test depends on the choice of statistic and on the population from which the observations come. Other statistics usable for two-sample comparisons as in Example 3.1:

- the sum $\sum_{i=1}^{n} Y_i$ of the observations of one treatment;
- the two-sample t-statistic $T = (\bar{Y}_n - \bar{X}_m)/\widehat{\text{se}}$, where $\widehat{\text{se}} = \sqrt{\frac{1}{m} + \frac{1}{n}}\, s_p$ and $s_p^2 = \{\sum_{i=1}^{m}(X_i - \bar{X}_m)^2 + \sum_{i=1}^{n}(Y_i - \bar{Y}_n)^2\}/(m+n-2)$;
- the difference of medians $\text{median}\{Y_i\} - \text{median}\{X_i\}$, useful for asymmetric distributions;
- the difference of trimmed means. E.g. the 10% trimmed mean is the mean of the middle 90% of the data after deleting the largest 5% and the smallest 5%; useful for symmetric distributions with outliers.

In the randomization test the observations $\{X_1, \ldots, X_m, Y_1, \ldots, Y_n\}$ are fixed, so the total $\sum_i^m X_i + \sum_i^n Y_i$ is the same under every assignment. Hence the statistic first considered,

$$D = \frac{1}{n}\sum_{i=1}^{n} Y_i - \frac{1}{m}\sum_{i=1}^{m} X_i = \frac{m+n}{mn}\sum_{i=1}^{n} Y_i - \frac{1}{m}\Big(\sum_{i=1}^{m} X_i + \sum_{i}^{n} Y_i\Big),$$

is an increasing function of $\sum_{i=1}^n Y_i$, and the randomization test with $\sum_{i=1}^n Y_i$ coincides with the one using D. The two-sample t-statistic T is also an increasing function of D, so the randomization p-value with T equals that with D. In contrast, using the difference of medians $\text{median}\{Y_i\} - \text{median}\{X_i\}$ can give a different result from D.

Proof that T is increasing in D (supplied). Let $\bar{Z}$ and $SST = \sum_i (X_i - \bar{Z})^2 + \sum_j (Y_j - \bar{Z})^2$ be the grand mean and total sum of squares of the pooled data; both are the same for every assignment. The ANOVA decomposition gives

$$SST = (m + n - 2)s_p^2 + \frac{mn}{m+n}D^2,$$

because the between-group sum of squares is $m(\bar{X}_m - \bar{Z})^2 + n(\bar{Y}_n - \bar{Z})^2 = \frac{mn}{m+n}(\bar{Y}_n - \bar{X}_m)^2$. Hence

$$T = \frac{D}{\sqrt{\frac{1}{m} + \frac{1}{n}}\sqrt{(SST - \frac{mn}{m+n}D^2)/(m+n-2)}} = g(D),$$

and $g(x) = cx/\sqrt{a - bx^2}$ ($a = SST$, $b = \frac{mn}{m+n}$, $c > 0$) satisfies $g'(x) = ca/(a - bx^2)^{3/2} > 0$ on $bx^2 < a$, where D always lies. So T is a strictly increasing function of D. ■

3.3 The population model

In Example 3.2, m flies were sampled at random from population 1 (forest tsetse flies) with antenna lengths $X_1, \ldots, X_m$, and n flies from population 2 (river tsetse flies) with $Y_1, \ldots, Y_n$. We want the two-sample t-statistic as test statistic. Since the observations were sampled at random from populations, the hypotheses can be stated through the distributions F of population 1 and G of population 2:

$$H_0 : F = G \quad \text{vs.} \quad H_1 : F \neq G. \tag{3.4}$$

In the population model the randomness of sampling from the population is the basis of inference. Random sampling yields $X_i \overset{\text{i.i.d.}}{\sim} F$, so the observations represent the population F well. Hence the results of tests and estimates in the population model can be claimed to hold for the whole population.[2] By contrast, inference in the randomization model is inference about the given data (the sample at hand).

If each population is normal with equal variances, $F \equiv N(\mu_1, \sigma^2)$ and $G \equiv N(\mu_2, \sigma^2)$, and the hypotheses become hypotheses about the location parameters of a parametric model:

$$H_0 : \mu_2 - \mu_1 = 0 \quad \text{vs.} \quad H_1 : \mu_2 - \mu_1 \neq 0. \tag{3.5}$$

Here the two-sample t-test is known to be the most efficient test, and the null distribution of the two-sample t-statistic is the t_{m+n-2} distribution. If normality cannot be assumed (i.e. for the hypotheses (3.4) rather than (3.5)), can we obtain the null distribution of t?

[2] In complete random sampling the observations are independent and each follows its population. In observational studies the observations are often not truly independent, or it is impossible to sample so that they exactly follow the population. Here we assume the samples are approximately independent and approximately follow the population (i.e. roughly satisfy the i.i.d. assumption). In the randomization model of Section 3.2, random assignment to the given units can be implemented exactly.

The answer is the **permutation test** in the population model. Its idea: if H_0 is true, the two populations F and G coincide, so the $N = m+n$ random variables all come from the same distribution, whichever population (F or G) they come from. So the observed sample $X_1, \ldots, X_m, Y_1, \ldots, Y_n$ is one of the $\binom{N}{m}$ possible partitions of these N values into population 1 (m values) and population 2 (n values). Given the values $\{X_1, \ldots, X_m, Y_1, \ldots, Y_n\}$, each of the $\binom{N}{m}$ partitions has the same probability $1/\binom{N}{m}$ of having occurred. The distribution of t over the $\binom{N}{m}$ partitions is the **permutation distribution** (shuffle distribution).[3] A test using it as the null distribution of t to compute p-values is called a **permutation test**.

To understand the permutation distribution, arrange the data as a table with the antenna length (variable `AL`) and a categorical variable (`label`) indicating forest or river fly. The different partitions (shuffles) in the permutation test amount to shuffling the `label` variable of the data (columns 1–2 are the original data). The third variable `permuted.label` in the R output is one of the $\binom{N}{m}$ values obtained by shuffling `label`, and under H_0 the original labels and `permuted.label` have the same probability.

```
   AL  label permuted.label
1  1.24 forest  river
2  1.38 forest  forest
3  1.36 forest  forest
4  1.40 forest  forest
5  1.38 forest  forest
6  1.48 forest  forest
7  1.38 forest  river
8  1.54 forest  forest
9  1.56 forest  forest
10 1.14 river   river
11 1.20 river   forest
12 1.30 river   forest
13 1.26 river   river
14 1.28 river   river
15 1.18 river   river
```

For the tsetse data of Example 3.2, let us obtain the null distribution (permutation distribution) of the test statistic. There are $\binom{15}{9} = 5005$ shuffles, so listing all of them by hand as in Example 3.1 is impractical. The `label` and `permuted.label` above are two of the 5005 cases; the two-sample t-statistic computed from each gives $t_{\text{obs}} = 4.08$ (from `label`) and $t_i = 2.81$ (from `permuted.label`). Doing this for all 5005 cases gives the permutation distribution (Figure 3.2 of the book; for R see Section 3.6.1). The two-sided p-value is $p = P(|T| \geq |t_{\text{obs}}| \mid H_0)$; in Example 3.2, 11 of the $|t_i|$ are at least $|t_{\text{obs}}| = 4.08$, so $p = 11/5005 \approx 0.002$ and the antenna lengths of the two tsetse species differ significantly.

3.3.1 Conditional inference

The permutation test in the population model is a **conditional inference** method. The permutation distribution is not the exact null distribution, but the null distribution under the condition that, when H_0 is true, the observed values (above, the values of `AL`) are exactly the values we observed. Moreover, whatever the population distribution, the permutation distribution given the observed values is distribution-free, because with the data values fixed it is the distribution of the statistic over the $\binom{N}{m}$ shuffles, each with the same probability (a discrete uniform distribution).

Note that the permutation test in the population model is exactly the same as the randomization

[3] The number of distinct values of the permutation distribution is $\binom{N}{m} = N!/(m!n!)$, the number of ways (combinations) of splitting N objects into m and n. This differs from the number of permutations of m out of N, ${}_NP_m = N!/m!$, and from the number $N!$ of orderings of all N objects. In permutation distributions and tests, "permutation" means shuffling, not ordered arrangement; the number of possible shuffles is $\binom{N}{m}$.

test in the randomization model. For this reason the permutation test, like the randomization test, is a **conditionally exact inference** (conditional because the data values are fixed). In formulas: if $\alpha = k/\binom{N}{m}$,

$$P(\text{reject } H_0 \mid H_0, \text{data}) = P(p \le \alpha \mid H_0, \text{data}) = \alpha.$$

That is, given the data the level is controlled exactly. Since it is controlled for every data set from the population,

$$P(\text{reject } H_0 \mid H_0) = E\{P(\text{reject } H_0 \mid H_0, \text{data})\} = E\{\alpha\} = \alpha,$$

so the level α is controlled unconditionally too.

3.3.2 The random permutation test

A permutation (randomization) test computes the statistic for all shuffles and obtains the exact null distribution. In Example 3.1 there were $\binom{7}{4} = 35$ shuffles, and in Example 3.2 $\binom{15}{9} = 5005$; the number of shuffles grows very fast with the sample size. With $(n, m) = (10, 10)$ it is about 6.7×10^{11}, too many for a personal computer.

A simple remedy is random sampling from the permutation distribution: recomputing the statistic for a few thousand randomly sampled shuffles takes little time on most computers. The p-value of this **random permutation test**, or Monte Carlo permutation test, is obtained as follows, for a test rejecting H_0 when $T_{\text{obs}} \ge c$ for a critical value c:

1. Choose the number M of random samples approximating the permutation distribution; $M = 999$ or $M = 1999$ is common.
2. Compute T_{obs} from the unshuffled data.
3. For $i = 1, \ldots, M$ generate a random shuffle splitting the N units into m and n, and compute T_i from the shuffled data.
4. The proportion of $\{T_{\text{obs}}, T_1, \ldots, T_M\}$ at least T_{obs} is the approximate p-value. With $T_0 = T_{\text{obs}}$,

$$\hat{p} = \frac{\sum_{i=0}^{M} I(T_i \ge T_{\text{obs}})}{M+1}.$$

In the third step we generate M independent shuffles splitting N into m and n, i.e. we sample M times from the $\binom{N}{m}$ shuffles with equal probability. To program this we either shuffle the category of each unit (in Example 3.2 the value of `label`) or shuffle the units' ids (in Example 3.2 the numbers `1:15` in the first column).

Shuffling the unit ids. Generate a random permutation of $\{1, \ldots, N\}$, assign the first m units to the first category and the remaining n to the second. E.g. for Example 3.1 with $N = 7$, $m = 4$, the following R code generates one random shuffle:

```
> N <- 7
> rand.permutation <- sample(1:N, N, replace = FALSE)
> rand.permutation
[1] 7 5 4 1 6 2 3
```

In this shuffle, units $7, 5, 4, 1$ are assigned to the first category and units $6, 2, 3$ to the second.

If the permutation $(5, 7, 4, 1, 6, 2, 3)$ happened to be generated, it differs from $(7, 5, 4, 1, 6, 2, 3)$ but has the same probability (each $1/N!$) and gives the same shuffle. A random permutation is one of the $N!$ permutations chosen at random, and $m!n!$ of them are equivalent to $(7, 5, 4, 1, 6, 2, 3)$ (give the same shuffle). So each distinct shuffle has probability $(m!n!)/N! = 1/\binom{N}{m}$, and shuffling unit ids by a random permutation is an exact way of producing a random shuffle.

Shuffling category values. The second way is to shuffle the values of the categories. If the categories before shuffling are `New,New,New,New,Old,Old,Old`, applying the permutation $(7, 5, 4, 1, 6, 2, 3)$ from the code above gives `Old,Old,New,New,Old,New,New`. (The first unit takes the category of the original 7th value, the second unit that of the 5th, and so on.) In this shuffle units 3, 4, 6, 7 are in the first category and units 1, 2, 5 in the second. Compared with shuffling ids, the same permutation gives a different shuffle, but shuffling the categories also uses a random permutation and is an exact way of producing an arbitrary shuffle. The following R code shows both ways of shuffling the category values with a random permutation:

```
> N = 7
> label = c("New","New","New","New","Old","Old","Old")
> set.seed(0)
> perm.id = sample(1:N, N, replace = FALSE)
> perm.id
[1] 6 1 4 3 7 2 5
> label[perm.id]
[1] "Old" "New" "New" "New" "Old" "New" "Old"
> set.seed(0)
> permuted.label = sample(label, N, replace = FALSE)
> permuted.label
[1] "Old" "New" "New" "New" "Old" "New" "Old"
```

Accuracy of the random permutation p-value. The random permutation p-value $\hat{p}$ approximates the permutation p-value p; more precisely it estimates p from M random samples, where M is the number of random shuffles. How far is $\hat{p}$ from the exact p? $\hat{p}$ is essentially the mean of the independent Bernoulli variables $I(T_i \geq T_{\text{obs}})$ with $E\{I(T_i \geq T_{\text{obs}})\} = P(T_i \geq T_{\text{obs}}) = p$, so

$$E(\hat{p}) = p, \qquad \text{se}(\hat{p}) = \sqrt{\frac{p(1-p)}{M}} \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{M}}.$$

With $M = 999$, $\hat{p} = 0.01$ has standard error 0.003 and $\hat{p} = 0.1$ has 0.009, so $\hat{p}$ is quite accurate.

3.4 How permutation tests work

Permutation tests are nonparametric tests needing very few assumptions about the population, and they are conditionally exact given the data. They can be schematized as follows:

1. State H_0 and H_1, and choose a test statistic D that tends to be large under H_1.
2. Check that the data are **exchangeable** under H_0, and determine the possible shuffles within the range of exchangeability.
3. If the number of shuffles is not large (e.g. $M = 2000$ or fewer), obtain for exact inference the distribution of the statistic over all shuffles (the re-randomized or permutation distribution),[4] i.e. recompute the statistic d_i for each shuffle.
4. The p-value is the proportion of the d_i at least the unshuffled value D_{obs}.

Whatever statistic is chosen in step 1, the permutation test controls the type I error exactly or conservatively. If the family of populations can be specified, choosing a statistic with higher power in that family is better. E.g. for a two-sample test with approximately normal populations, choose the two-sample t-statistic (or the difference of means). If the distribution looks skewed or has outliers, a difference of medians or trimmed means raises the power. Replacing observations by

[4] If there are too many shuffles, estimate the null distribution by the random permutation test.

special scores such as ranks or signs and computing the statistic from them often reproduces classical nonparametric methods.

Permutation tests need very few assumptions about the population; the needed assumption is essentially the **exchangeability of the data under the null hypothesis** described in step 2. Exchangeability is the key assumption that allows the null distribution to be derived in a permutation test. Writing a reordering of n observations as a (one-to-one) permutation function $\pi : \{1, \ldots, n\} \to \{1, \ldots, n\}$, exchangeability is defined as follows.[5]

Definition 3.1 (Exchangeable random variables)

Let $F(x_1, \ldots, x_n)$ be the joint distribution of random variables $X_1, \ldots, X_n$. The random variables are **exchangeable** if for every permutation $\pi : \{1, \ldots, n\} \to \{1, \ldots, n\}$ of their order, the joint distribution of $X_{\pi(1)}, \ldots, X_{\pi(n)}$ is also $F(x_1, \ldots, x_n)$.

Exchangeability is a very weak assumption. E.g. independent identically distributed (i.i.d.) random variables are exchangeable whatever their distribution. Even identically distributed but dependent random variables can be exchangeable. So exchangeability is a very weak nonparametric assumption.

Why exchangeability makes the permutation test valid (supplied)

Let $Z = (Z_1, \ldots, Z_N)$ be exchangeable under H_0 and $D = D(Z)$ a statistic. Let $\{\pi_1, \ldots, \pi_K\}$ be the $K = N!$ permutations, and $p(Z) = \frac{1}{K} \sum_k I(D(Z_{\pi_k}) \geq D(Z))$ the permutation p-value. Then $P(p(Z) \leq \alpha \mid H_0) \leq \alpha$.

Proof. By exchangeability $Z_{\pi_k} \overset{d}{=} Z$ for each k, so for U uniform on $\{1, \ldots, K\}$ independent of Z, $Z_{\pi_U} \overset{d}{=} Z$. Given the multiset $\{D(Z_{\pi_k})\}_k$ (which is the same for Z and every Z_{π_U}, as the permutations form a group), $D(Z_{\pi_U})$ is a uniformly chosen element of it. The p-value is the proportion of elements at least the chosen one, and a uniformly chosen element is among the top j (counted with ties) with probability at most j/K. So $P(p \leq \alpha) \leq \alpha$, with equality when αK is an integer and there are no ties. Restricting to the $\binom{N}{m}$ two-sample splits (each realized by $m!n!$ permutations) gives the same statement.

Some examples of exchangeability of random variables (or data):

Example 3.5 (i.i.d. random variables)

In a two-population model, observations $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} G$ are independent, each by simple random sampling. If $H_0 : F = G$ is true, all observations are independent with the same distribution ($F = G$), so they are exchangeable. Likewise, observations i.i.d. in a one-population model are exchangeable, whatever the population and whether or not H_0 holds.

Example 3.6 (Compound symmetry)

To assess elementary pupils' achievement, first choose an elementary school in Seoul, then randomly sample n upper-grade pupils of that school and observe their common achievement test scores $X_1, \ldots, X_n$. Assume the scores are normal and, since all pupils in the sample attend the same school, strongly positively correlated: for all $i \neq j$, $X_i \sim N(\mu, \sigma^2)$ and $\text{Corr}(X_i, X_j) = \rho > 0$. Then the joint distribution of $(X_1, \ldots, X_n)$ is multivariate normal

[5] A permutation function π is a bijection of a set onto itself. E.g. on $\{1, 2, 3\}$, $\pi(1) = 2$, $\pi(2) = 1$, $\pi(3) = 3$ swaps the first two elements 1 and 2.

$N_n(M, \Sigma)$ with mean and covariance

$$M = \begin{pmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{pmatrix}, \qquad \Sigma = \sigma^2 \begin{pmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & & \ddots & \\ \rho & \rho & \cdots & 1 \end{pmatrix}.$$

Reordering the variables does not change the mean or the covariance, so $(X_1, \ldots, X_n)$ is exchangeable but not independent. Covariance matrices of this form arise often in cluster samples; this is the *compound symmetry* structure.

The next example is a case where the data are not exchangeable even under the null hypothesis.

Example 3.7

In a two-population model, $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} G$ are independent, each a simple random sample, with $F = N(\mu_1, 1)$ and $G = N(\mu_2, 4)$; we test $H_0 : \mu_1 = \mu_2$. Even under H_0, F and G differ, so the joint distributions of $(X_1, X_2, \ldots)$ and $(X_1, Y_1, \ldots)$ differ: these observations are not exchangeable.

Applying a permutation test to data that are not exchangeable under H_0 may fail to control the type I error exactly, so in principle permutation tests cannot be applied.

3.5 Advantages and disadvantages of permutation tests

Most tests in Part I of this book can be understood as permutation tests. Their advantages:

1. They apply to any data satisfying exchangeability: the assumptions on the population can be very weak, so they are nonparametric methods applicable to very broad families.
2. There is no restriction on the choice of test statistic.
3. The null distribution of any statistic is very easy to obtain.

Disadvantages:

1. Computation takes longer than for parametric methods and traditional (rank- and sign-based) nonparametric procedures.
2. Because much computation is needed, they are hard to use for inference other than testing via p-values (e.g. for confidence intervals).
3. With shuffles that do not satisfy exchangeability, the type I error may not be controlled.

3.6 Various permutation tests

3.6.1 Two-sample permutation tests

To introduce permutation tests, Section 3.1 looked at randomization or permutation tests in the randomization model (two treatments) or the population model (two populations). Both use the same test procedure, so both are called the two-sample permutation test. With F the distribution of the first population and G that of the second, the null hypothesis is

$$H_0 : F = G, \tag{3.6}$$

and three alternatives can be considered (Figure 3.3 of the book):

- $H_1 : F(x) \leq G(x)$ for all x: "observations of population 1 tend to be larger than those of population 2."
- $H_1 : F(x) \geq G(x)$ for all x: "observations of population 1 tend to be smaller than those of population 2."
- $H_1 : F(x) \neq G(x)$: "the two distributions differ (in location, in variance, or in shape)."

If the two populations are equal under H_0 and differ only in the location parameter under H_1, the location difference Δ can be used as the parameter. Then

$$F(x) = G(x + \Delta), \tag{3.7}$$

or in terms of densities $f(x) = g(x + \Delta)$, so for $X \sim F$, $Y \sim G$,

$$P(X \leq x) = P(Y \leq x + \Delta), \tag{3.8}$$

i.e. $X + \Delta \overset{d}{=} Y$ or $X \overset{d}{=} Y - \Delta$: the second population is the first one shifted by Δ (to the right). Then the null hypothesis (3.6) is simply a statement about the parameter Δ, $H_0 : \Delta = 0$, and the three alternatives become lower-tailed, upper-tailed and two-sided:

- (lower-tailed) $H_1 : \Delta < 0$: "population 2 lies to the left of population 1";
- (upper-tailed) $H_1 : \Delta > 0$: "population 2 lies to the right of population 1";
- (two-sided) $H_1 : \Delta \neq 0$.

A simple estimator of Δ, or one of the following, as test statistic D tends to take different values under the null and the alternative:

- the sum of the observations of the new treatment $\sum_{i=1}^{n} Y_i$;
- the two-sample t-statistic $t = (\bar{Y}_n - \bar{X}_m)/\widehat{\text{se}}$;
- the difference of medians or of trimmed means.

With m observations from the first population, n from the second, and $N = m + n$, the null distribution of D is obtained from all $\binom{N}{m}$ shuffles or approximated by random permutations. If $D = \bar{X} - \bar{Y}$ is used and $d_1, \ldots, d_M$ are the values from random permutations, the p-values are as follows (write d_0 for the observed D_{obs}):

- lower-tailed: $p_l = \sum_{i=0}^{M} I(d_i \leq D_{\text{obs}})/(M + 1)$;
- upper-tailed: $p_u = \sum_{i=0}^{M} I(d_i \geq D_{\text{obs}})/(M + 1)$;
- two-sided: $p = \sum_{i=0}^{M} I(|d_i| \geq |D_{\text{obs}}|)/(M + 1)$.

Generating all shuffles in the two-sample test. To generate the null distribution of a statistic D in a two-sample test from all shuffles, all $\binom{N}{m}$ cases must be generated. The R code below is an example. (Note: for large N this approach is not recommended.)

```
> dat = read.csv("data/tsetse.csv"); N = 15; m = 9; n = 6
> head(dat,3)
    AL  label
1 1.24 forest
2 1.38 forest
3 1.36 forest
```

First fix the total sample size N and the size m of the first sample. To shuffle the unit ids we use `gtools::combinations()`.

```
> allperms <- gtools::combinations(N,m)
> head(allperms,3)
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9]
[1,]    1    2    3    4    5    6    7    8    9
[2,]    1    2    3    4    5    6    7    8   10
[3,]    1    2    3    4    5    6    7    8   11
> nrow(allperms)
[1] 5005
```

Each row of `allperms` holds the $m = 9$ ids of the first sample after shuffling, and there are $\binom{N}{m} = 5005$ distinct rows. With $D = \bar{Y}_n - \bar{X}_m$ the null distribution is generated by

```
> D.perm = vector(length = choose(N,m))
> for (i in 1:choose(N,m)){
+   permuted.x = dat$AL[allperms[i,]]
+   permuted.y = dat$AL[-allperms[i,]]
+   D.perm[i] = mean(permuted.y) - mean(permuted.x)
+ }
> head(D.perm,3)
[1] -0.18666667 -0.07000000 -0.08666667
```

Instead of the `for` loop one can use `apply` for speed:

```
> D.perm = apply(allperms,1,
+               function (perm.x.id){
+                 permuted.x = dat$AL[perm.x.id]
+                 permuted.y = dat$AL[-perm.x.id]
+                 mean(permuted.y) - mean(permuted.x)
+               } )
> head(D.perm,3)
[1] -0.18666667 -0.07000000 -0.08666667
```

In any case the "exact" two-sided p-value (null hypothesis (3.6)) is

```
> Dobs = with(dat,
+   mean(AL[label == "river"]) - mean(AL[label == "forest"])
+ )
> (p = mean(abs(D.perm) >= abs(Dobs)))
[1] 0.002197802
```

Random permutation test for two samples. A random permutation test needs no enumeration of cases and is simpler to implement. The R code below computes the lower-tailed p-value with the observed D_{obs} above by random permutations. There is no need to compute the statistic for each case of the shuffles; generate random permutations with `sample()` and repeat (`replicate()`) the computation M times.

```
> M = 1999
> D.perm = replicate(M, {
+   x.id = sample(1:N,m, replace =FALSE)
+   permuted.x = dat$AL[x.id]
+   permuted.y = dat$AL[-x.id]
+   mean(permuted.y) - mean(permuted.x)
+ })
> (p = mean(c(Dobs,D.perm) <= Dobs))
[1] 5e-04
```

3.6.2 One-sample permutation tests

For a test about the location parameter of one population from a sample, let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$. We compare $H_0 : \theta = \theta_0$ for the median $\theta = Q_F(0.5)$ of F with one of

$$H_1 : \theta > \theta_0, \qquad H_1 : \theta < \theta_0, \qquad H_1 : \theta \neq \theta_0.$$

The test statistic can be one of:

- the t-statistic $T = \frac{\bar{X}_n - \theta_0}{s/\sqrt{n}}$, with $s^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ the sample variance;
- the sign statistic $S = \sum_{i=1}^n I(S_i = +) = \sum_{i=1}^n I(X_i > \theta_0)$;
- Wilcoxon's signed-rank statistic $SR_+ = \sum_{i=1}^n I(X_i > \theta_0) \cdot R_i$.

With the statistic chosen, the first of the four steps of Section 3.4 is done. What, then, is the permutation distribution (null distribution under H_0) of the statistic?

First consider the sign statistic S. Under H_0, an observation X_i is above the median θ_0 with the same probability as below it ($P(X_i > \theta_0) = P(X_i < \theta_0) = 1/2$), so shuffling the signs of the n observations leaves the distribution unchanged: the joint distribution of $(S_1, \ldots, S_n)$ equals that of $(-S_1, S_2, \ldots, S_n)$ (the joint distribution is the same however many "$-$" signs are attached and wherever). So the permutation distribution here is obtained not by shuffling the order of observations but by flipping their signs.[6] There are $M = 2^n$ possible sign flips, through which the permutation distribution of S is obtained. It does not depend on the observed values (only on the signs S_i), and coincides with the null distribution $B(n, 1/2)$ of S derived when the sign test was introduced.

For the permutation distribution of Wilcoxon's SR_+, recall that each observation's contribution to SR_+ splits into a rank R_i and a sign S_i. If F is symmetric about θ_0, an observation X_i with rank R_i is equally likely to lie left or right of θ_0. That is, the joint distribution of $((S_1, R_1), \ldots, (S_n, R_n))$ given $(R_1, \ldots, R_n)$ equals that of $((-S_1, R_1), \ldots, (S_n, R_n))$, however many "$-$" signs are attached and wherever. When the population is symmetric about θ_0, the conditional null distribution (permutation distribution) of $X_1, \ldots, X_n$ given the ranks is obtained by flipping the signs of the observations, and this is Wilcoxon's signed-rank distribution.

Summarizing, the sign test and the Wilcoxon signed-rank test are permutation tests with statistics S and SR_+. Also, the permutation distribution of SR_+ cannot be obtained when the population does not satisfy the symmetry assumption: with the ranks R_i fixed, the distribution of S_i changes with the sign flips, violating exchangeability, so re-randomization cannot be performed.

Next consider the permutation distribution of the t-statistic. In a permutation test the t-statistic can be used even if normality is violated, but the population must be assumed symmetric. Compare each X_i with the hypothesized median θ_0: with $Z_i = X_i - \theta_0$ and sign $S_i = +I(X_i > \theta_0) - I(X_i < \theta_0)$,

$$X_i - \theta_0 = S_i |Z_i|.$$

Given the observed $(|Z_1|, \ldots, |Z_n|)$, the conditional distributions of S_i and $-S_i$ coincide,

$$P(S_i = \text{“}+\text{”} \mid |Z_i|) = P(S_i = \text{“}-\text{”} \mid |Z_i|) = 1/2,$$

so, as for the permutation distribution of SR_+, the permutation distribution of T is obtained using the observations shuffled by the $M = 2^n$ distinct sign flips,

$$X_i^{\text{perm}} = \theta_0 \pm S_i |Z_i|.$$

[7]

[6] Again, in this case the shuffles differ from permutations, which is common. That is why "shuffle distribution" and "shuffle test" are more precise terms than permutation distribution and permutation test.

[7] For reference, the one-sample t-test and the one-sample permutation test with the t-statistic have very similar efficiency (Helwig, 2019).

3.6.3 Testing independence for categorical data

The data of the next example are categorical.

Example 3.8

Is excessive salt intake related to cardiovascular disease? The data below record whether 30 cardiovascular patients and 30 people of similar health without the disease consume too much salt:

	high salt	low salt	total
cardiovascular disease	5	25	30
no disease	2	28	30
total	7	53	60

We want to test with these data whether excessive salt intake and cardiovascular disease are related. One may compare the proportion of excessive salt takers between the populations "disease" and "no disease," or the proportion of patients between the populations "high salt" and "low salt." The populations and parameters considered differ with how the data were sampled, but the test procedure is the same. Concretely, let p_1 be the proportion of patients in the high-salt group and p_2 that in the low-salt group, and test

$$H_0 : p_1 = p_2 \quad \text{vs} \quad H_1 : p_1 > p_2.$$

(The alternative may also be two-sided.) In categorical data analysis this is called a **test of homogeneity**, or a **test of independence** if the data are a random sample, and the χ^2 test can be used; but it is accurate only for large samples and cannot be applied here. Instead, a permutation test with a simple statistic can be used. Let the statistic D be the number of patients who take too much salt and have the disease (under the alternative D tends to be large). The observed value is $D_{\text{obs}} = 5$.

Rearranged as a data table (rows for the $n = 60$ patients, two columns for the two variables), with `Cardio = Yes` for disease and `Salt = H` for excessive intake:

```
   Cardio Salt
1      No    H
2      No    H
3     Yes    H
4     Yes    H
5     Yes    H
6     Yes    H
7     Yes    H
8      No    L
9      No    L
10     No    L
...
```

If $H_0 : p_1 = p_2$ is true, the probability of disease is the same whether or not one is in the high-salt group (the probability of `Cardio = Yes` is the same for `Salt = H` and `Salt = L`). We do not know this probability $p = p_1 = p_2$ exactly. Fisher's exact test for two categorical variables fixes that half of the `Cardio` observations (30 of 60) are `Yes` and that 7 values of `Salt` are `Yes`, and considers all shuffles possible under H_0. Such shuffles fix the values of one variable (e.g. `Cardio`) and shuffle the values of the other (`Salt`). Of the $N = 60$ observations, $k = 7$ have `Salt = H`, so there are $\binom{60}{7} = 386206920$ distinct shuffles, and all of them give the exact permutation distribution of D. (Using it as the null distribution gives the p-value of Fisher's exact test.)

The fixed variable `Cardio` has only two possible values, so the number of possible values of D (8 here) is much smaller than the number of shuffles (about 3.8×10^8). So it is not necessary to

consider all shuffles to obtain the permutation distribution. Fisher's exact test could be used from the 1930s because this permutation distribution is a well-known distribution (the hypergeometric distribution[8]).

Example 3.9 (Example 3.8 continued)

Compare the observed $D_{\text{obs}} = 5$ with the null distribution of D to get the p-value. D can take the values $d = 0, 1, \ldots, 7$, and $P(D = d)$ can be obtained from the hypergeometric distribution (or as the proportion of all shuffles giving $D = d$). The R code below extracts the hypergeometric probabilities directly to get the null distribution of D:

```
> nulldist = dhyper(0:7, m = 30, n = 30, k = 7)
> round(nulldist,2)
[1] 0.01 0.05 0.16 0.29 0.29 0.16 0.05 0.01
```

The alternative is upper-tailed, so the p-value is

$$P(D \geq D_{\text{obs}}) = P(D \geq 5) = P(D = 5) + P(D = 6) + P(D = 7) = 0.211,$$

not enough evidence to reject H_0.

For reference, writing the two variables as C_i, S_i, under H_0 the joint distribution of $((C_1, S_1), (C_2, S_2))$ equals that of $((C_2, S_1), (C_1, S_2))$. So (C_i) is exchangeable and the condition of the permutation test is satisfied.

[8] When m of N patients have the disease and $N - m$ do not, and k of them are chosen at random and assigned to the high-salt group, the number X of patients with disease assigned to the high-salt group follows the hypergeometric distribution with probability mass function $f(x) = P(X = x) = \binom{m}{x}\binom{N-m}{k-x}\Big/\binom{N}{k}$. In Example 3.8, $N = 60$, $m = 30$, $k = 7$.